

Cyclic Redundancy Check (CRC)

→ frame check sequence (FCS)

→ Polynomial function, when coefficients are either zero or one

Example:-

function:- $x^5 + x^2 + 1$

$G(x)$: Generator Polynomial function.

→ $(n+1)$ terms $[x^n \text{ to } x^0]$

→ Degree $[G(x)] = n$

→ Coefficient of term x^0 should be "one"
[$G(x)$ should not be divisible by x]
Completely

→ Coefficient of the term x^n should be one

→ Both transmitter and receiver must agree on same $G(x)$.

$$G(x) = x^n + \dots + 1$$

* Divisor:-

$$G(x) = x^n + \dots + 1$$

Divisor: binary string, $(n+1)$ bits $[1 \dots 1]$

Example:-

$$G(x) = x^3 + x^2 + 1$$

$$1 * x^3 + 1 * x^2 + 0 * x^1 + 1 * x^0$$

$$\text{Divisor} = 1101$$

* Message polynomial:-

$M(x)$: Message polynomial function

→ m terms, $[x^{(m-1)} \text{ to } x^0]$

→ Coefficients are either zero or one

Data (Message): Binary string (m -bits).

Example:-

$$M(x) = x^7 + x^4 + x^3 + x^0$$

$$1 * x^7 + 0 * x^6 + 0 * x^5 + 1 * x^4 + 1 * x^3 + 0 * x^2 + 1 * x + 0 * x^0$$

$$\text{Data} = 10011010$$

* Transmitter protocol :-

$$[M(x) * x^n] \text{ [Modulo-2 Division] } [G(x)]$$

[6-bit-wise X-OR]

Example :- $G(x) = x^3 + x^2 + 1$ ↗ Divisor

$$M(x) = x^7 + x^4 + x^3 + x$$

$$M(x) * x^3 = x^{10} + x^7 + x^6 + x^4 \quad \text{Divident}$$

n: degree of $[G(x)] = 3$

$$x^7 + x^6 + x^5 + x^4 + x^3 + 1$$

$$x^3 + x^2 + 1 \overline{) x^{10} + x^7 + x^6 + x^4}$$

$$x^{10} + x^9 + x^7$$

$$x^9 + x^6 + x^4$$

$$x^9 + x^8 + x^6$$

$$x^8 + x^4$$

$$x^8 + x^7 + x^5$$

$$x^7 + x^5 + x^4$$

$$x^7 + x^6 + x^4$$

$$x^5 + x^4$$

$$x^6 + x^5 + x^3$$

$$x^3$$

$$x^3 + x^2 + 1$$

$$x^2 + 1$$

Remainder Polynomial

$R(x)$:- Remainder Polynomial function

→ n terms, $[x^{n-1} \text{ to } x^0]$

→ Coefficient are either zero or one.

CRC :- (Remainder) :- Binary string (n-bit)

Example → 1

$$G(x) = x^3 + x^2 + 1$$

$$M(x) = x^7 + x^4 + x^3 + x$$

$$M(x) * x^3 = x^{10} + x^7 + x^6 + x^4$$

$$[M(x) * x^3] \text{ [Modulo-2 Division] } [G(x)]$$

$$R(x) = 1 * x^2 + 0 * x^1 + 1 * x^0$$

Transmitter Transmit

$$[M(x) * x^n] + [R(x)]$$

Example :-

→ $M(x) * x^3 = x^{10} + x^7 + x^6 + x^4$

$$R(x) = x^2 + 1$$

Transmitter Transmit → $x^{10} + x^7 + x^6 + x^4 + x^2 + 1$

Divisor = 1101
 Div = 10011010

$$M(x) \cdot x^3 = 10011010000$$

$$\begin{array}{r}
 1101 \overline{) 10011010000} \\
 \underline{1101} \\
 1001 \\
 \underline{1101} \\
 1001 \\
 \underline{1101} \\
 1001 \\
 \underline{1101} \\
 1001 \\
 \underline{1101} \\
 1001 \\
 \underline{1101} \\
 0
 \end{array}$$

$$\begin{array}{r}
 0010 \\
 0000 \\
 \hline
 0100 \\
 0000 \\
 \hline
 1000 \\
 1101 \\
 \hline
 101
 \end{array}$$

10011010101
 Data CRC

Consider the message $M = 101000101$. The cyclic redundancy check (CRC) for this message using the divisor polynomial $x^5 + x^4 + x^2 + 1$

AS 0110
 B 0101
 C 1010
 d 10110

The message 11001001 is to be transmitted using the CRC polynomial $x^3 + 1$ to protect it from errors. The message that should be transmitted is: —

a 11001001000
 b 11001001011
 c 11001010
 d 110010010011

A computer network uses polynomial over GF(2) for error checking with 8 bits as information bits and uses $x^3 + x + 1$ as the generator polynomial to generate the check bits in this network, the message 01011011 is transmitted as.

a 01011011010
 b 01011011011
 c 01011011101
 d 01011011100

* Revised Part 1 :-

$$[M(x) * x^n + R(x)] \text{ Modulo } -2 \text{ Division} [G(x)]$$

$$[M(x) * x^n + R(x)] \text{ Modulo-2 Division } [G(x)]$$

$R'(x) =$ Remainder of receiver (of above equation)

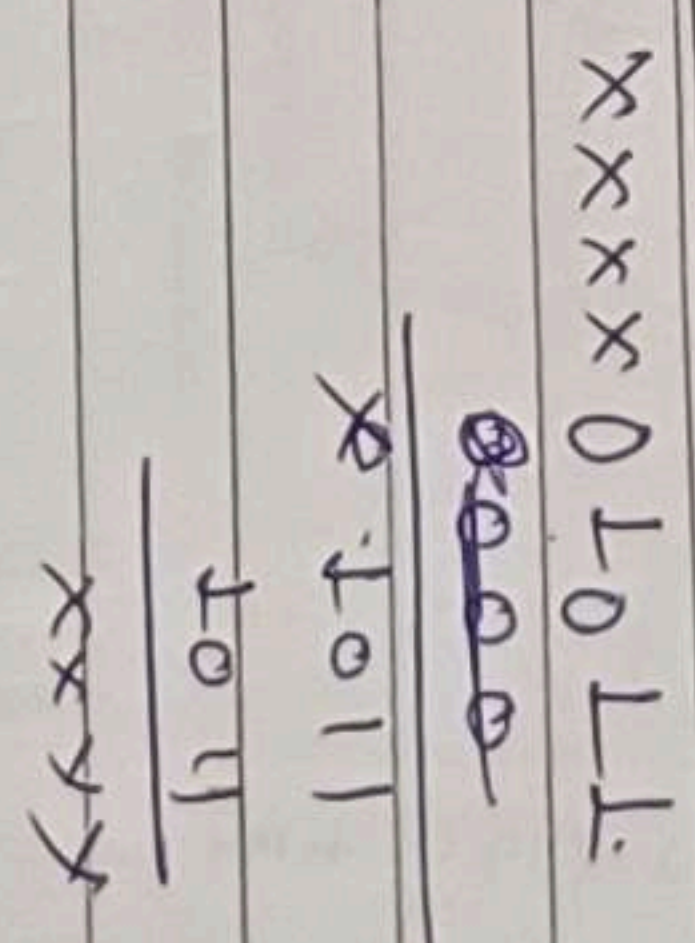
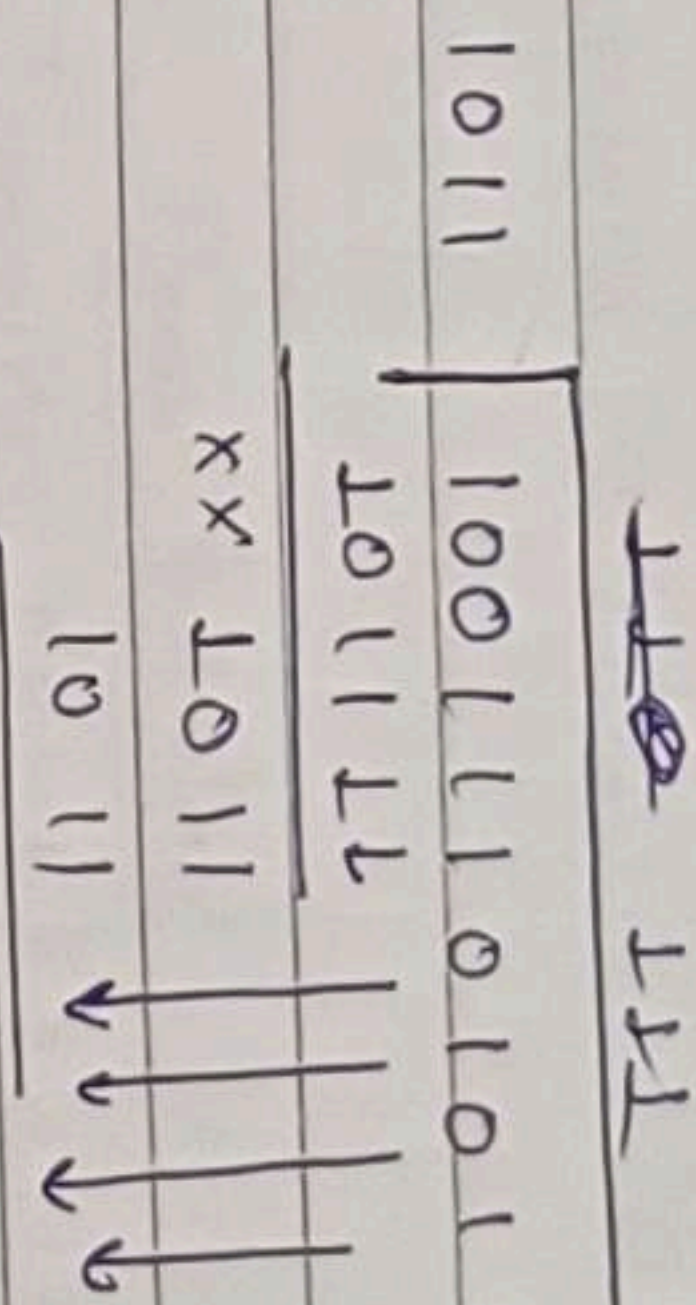
$R'(x) =$ Remainder of receiver (of above equation)

If $R'(x) = 0$

than Receiver concluded "No any error detected"

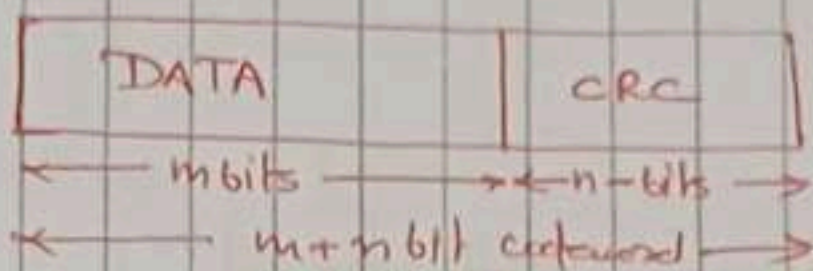
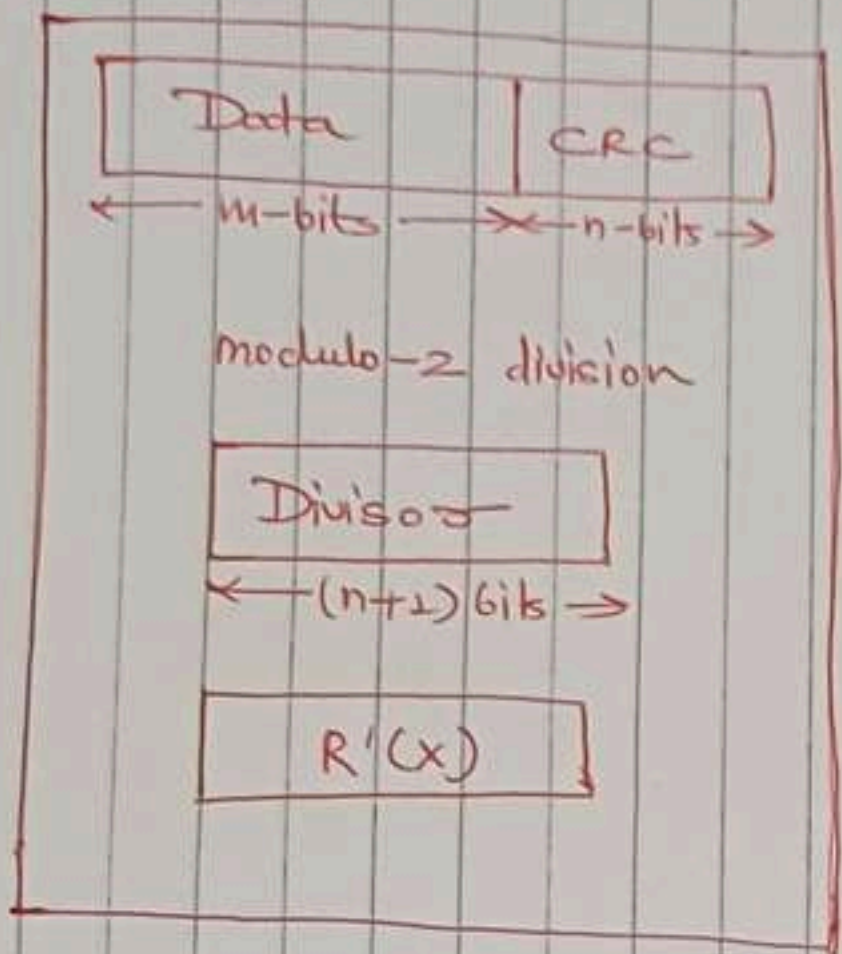
obs

Receiver concluded "Error detected"

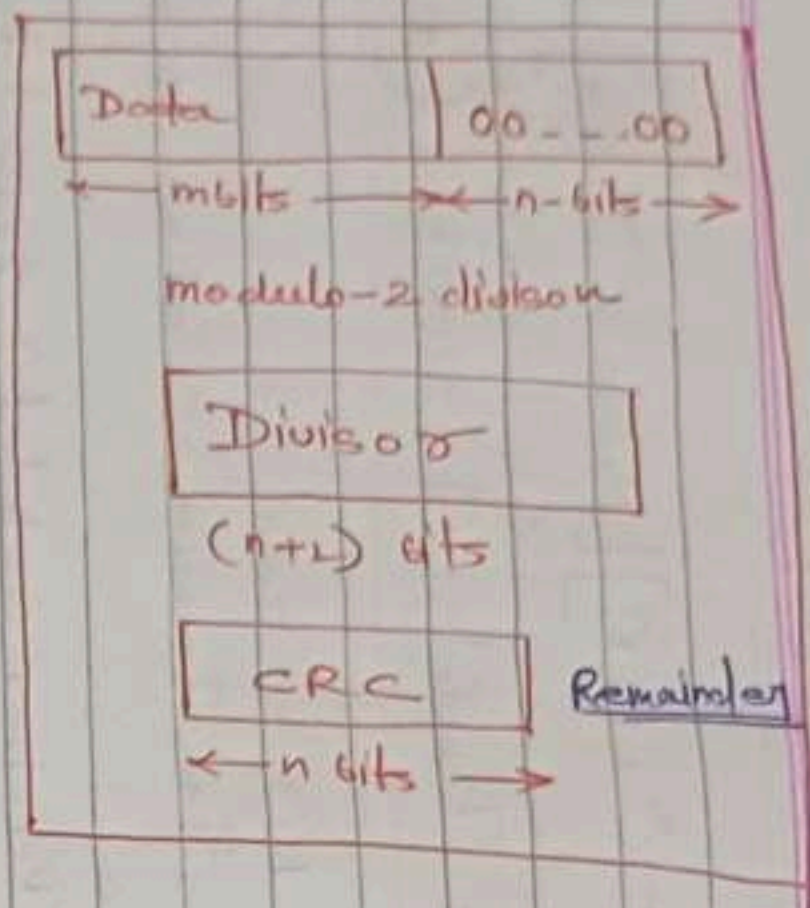


$$G(X) = X^n + \dots + 1 \text{ where } n > 0$$

Receiver



Sender / Transmitter



Case-1 No any error

Transmitter: $[M(x) * X^n] + [R(x)]$
Receiver Received: $[M(x) * X^n] + [R(x)]$

Receiver Protocol: $[M(x) * X^n + R(x)] \text{ [Modulo-2 Division]} [C(x)]$

Above equation will definitely lead "zero remainder"
Receiver conclude: "No any error detected"

Case-2 Error included

Transmitter transmit: $[M(x) * X^n] + [R(x)]$
Receiver received: $[M(x) * X^n] + [R(x)] + [E(x)]$

Error Polynomial

$E(x)$: Error polynomial function.
→ Coefficient here either 0 or 1

Data: m bits CRC: n bits

Coded word: $(m+n)$ bits

Degree $(E(x)) < (m+n)$

* for single bit error :-

$$E(x) = x^i$$

where $i=0$ to $(m+n-1)$.

Example :-

Transmitter transmitted :-

$$x^{10} + x^7 + x^6 + x^5 + x^3 + x + 1$$

$$10011101011$$

Receiver received :-

$$x^{10} + x^7 + x^6 + x^5 + x + 1$$

$$10011100011$$

$$E(x) = x^3$$

Example :-

Transmitter transmitted :-

$$x^{10} + x^7 + x^6 + x^5 + x^3 + x + 1$$

Receiver received :-

$$x^{10} + x^7 + x^6 + x^5 + x^4 + x^3 + x + 1$$

$$E(x) = x^4$$

* for two bit error

$$E(x) = (x^i + x^j)$$

where i, j are 0 to $(m+n-1)$ and $i \neq j$

Example :-

Transmitter transmitted :-

$$x^{10} + x^7 + x^6 + x^5 + x^3 + x + 1$$

Receiver received :-

$$x^{10} + x^9 + x^7 + x^6 + x^5 + x^3 + 1$$

$$E(x) = x^9 + x$$

Error included

Receiver protocol

$$[M(x) * x^n + R(x) + E(x)] \text{ Modulo-2 Division } [G(x)]$$

$$[M(x) * x^n + R(x)] \text{ Modulo-2 Division } +$$

$$[E(x)] \text{ Modulo-2 Division } [G(x)]$$

→ when this lead non-zero remainder

→ $G(x)$ have $(n+1)$ terms (length)

→ If $E(x)$ cause almost n length burst error then above equation always lead non-zero remainder.

All single bit error : $E(x) = x^i$

$$\frac{E(x)}{G(x)} = \frac{x^i}{x^n + \dots + 1} \quad 0 \leq i \leq (n-1)$$

~~Suppose~~

Suppose $G(x) = x^3 + 1$

$$E(x) = x^6 + x^3$$

$$\frac{E(x)}{G(x)} = \frac{x^6 + x^3}{x^3 + 1} = \frac{x^3(x^3 + 1)}{x^3 + 1}$$

$$G(x) = x^3 + 1$$

$$E(x) = x^5 + x^3$$

$$E(x) = \frac{x^5 + x^3}{x^3 + 1}$$

* CRC Property :-

→ CRC can detect any length burst error up to the degree of generator polynomial

$$\text{degree}[G(x)] = n$$

→ for odd number of error $E(x)$ must have odd number of terms

→ if $(x+1)$ is a factor of $G(x)$ then CRC can detect "all odd number of error"

→ " $(x+1)$ is a factor of $G(x)$ " mean " $G(x)$ is completely divisible by $(x+1)$ " and " $G(x)$ has even number of terms."

$$G(x) = F_1(x) * F_2(x)$$

$$G(x) = (x+1) * F_2(x)$$

→ If $(x+1)$ is not a factor of $G(x)$ then CRC always detect "all even number of error"

Q $G(x)$ be the generator polynomial used for CRC checking. What is the condition that should be satisfied by $G(x)$ to detect odd number of bits in error.

as $G(x)$ contains more than two terms
 $\hookrightarrow G(x)$ does not divide $1+x^k$, for any k not exceeding the frame length
 $\hookrightarrow 1+x$ is a factor of $G(x)$
 $\hookrightarrow G(x)$ has an odd number of terms.

$\rightarrow$ for two bit error, $E(x)$ must have two terms only.
 $E(x) = (x^i + x^j)$
 where i, j are 0 to $(m+n-1)$.

$$E(x) = x^i (x^{j-i} + 1)$$

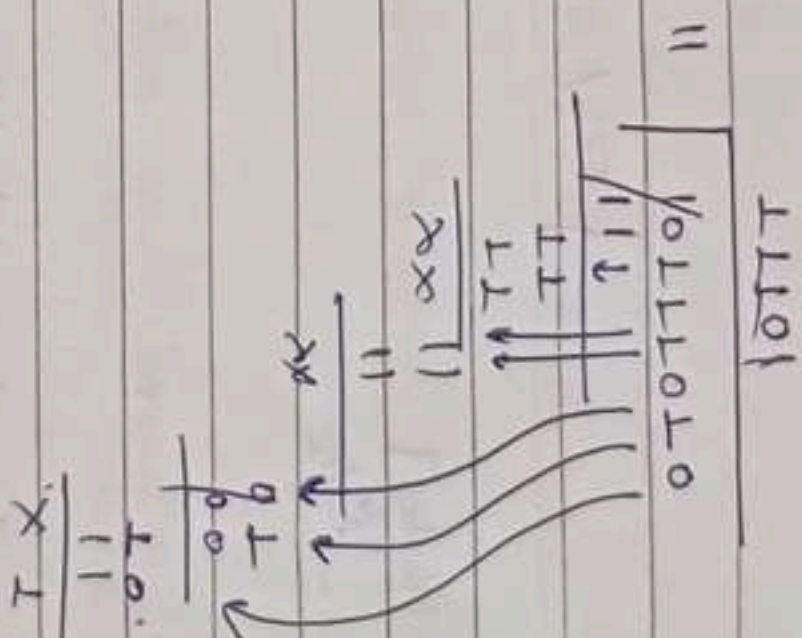
$\rightarrow$ To detect any two bit error
 $G(x)$ does not divide by $(x^k + 1)$
 $[\text{for any } k \text{ from } 2 \text{ to } (m+n-1)]$

Example :-

$$G(x) = x + 1$$

$$\text{Message (DATA)} = 1011101$$

$$\text{CRC} = 1$$



One-bit CRC [CRC-1] is same as one-bit parity with "even parity".

Example 2

$$G(x) = x^3 + 1$$

$$\text{Message} = 1011001$$

$$\text{CRC} = 000$$

Example 3

$$G(x) = x^3 + x^2 + x + 1$$

$$\text{Message (DATA)} = 11110001$$

$$\text{CRC} = 111$$

CRC can be "all zero bits" and can be "all one bits"